

SWE573 - Software Development Practice

Project Development Log - Local Craft Discovery Guide
Iteration 1: Repository, Git workflow, labels, and requirements elicitation

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Purpose

This document records the hands-on work completed while setting up and learning the project repository. It is intentionally process-oriented: commands were executed locally, their outputs were inspected, GitHub features were configured, and project decisions were documented as issues. The goal is to preserve evidence of the learning process rather than only the final repository state.

1. Project concept

The course domain is indigenous and handmade crafts. Several possible application directions were considered: a conventional information/blog site, a general map-based craft application, and a focused discovery guide. The selected concept is Local Craft Discovery Guide, a location-based application for travelers who want to discover culturally significant local crafts, understand their background, find where they can be experienced or purchased, and use ratings/reviews.

The concept takes inspiration from the focused discovery model of the Michelin Guide, but applies it to local crafts rather than restaurants and cuisine.

2. Repository setup and README

The GitHub repository was cloned locally and opened as a working Git repository. An initial mistake also became a useful lesson: running `git status` from the parent folder returned “not a git repository”; changing into the cloned repository directory resolved it.

```
git clone https://github.com/AnilBASARAN/SWE573-SoftwareDevelopmentAssignment.git
cd SWE573-SoftwareDevelopmentAssignment
git status
```

The README was expanded from a two-line placeholder into a project description containing the project concept, problem, target users, and project goals.

3. Git workflow learned through two README commits

Before staging changes, `git diff` was used to inspect exactly what had changed. This made the working-directory state visible before creating a commit.

```
git diff
git add README.md
git status
git commit -m "Add initial project description to README"
git log --oneline
```

The first README commit produced the short commit hash 0587d5d. A second README change added project goals and produced commit 9fdeb7d. The original repository commit remained visible

as a32afbe Initial commit.

```
9fdeb7d (HEAD -> main) Add project goals to README
0587d5d (origin/main, origin/HEAD) Add initial project description to README
a32afbe Initial commit
```

This output demonstrated an important Git concept: HEAD -> main represented the local branch position, while origin/main represented the locally known position of the remote GitHub branch. After committing but before pushing, local main was one commit ahead of origin/main.

```
git status
Your branch is ahead of 'origin/main' by 1 commit.
```

After pushing, both references pointed to the same commit:

```
git push origin main

9fdeb7d (HEAD -> main, origin/main, origin/HEAD) Add project goals to README
0587d5d Add initial project description to README
a32afbe Initial commit
```

The practical workflow learned in this iteration was: edit -> git diff -> git add -> git commit -> git push.

4. GitHub repository structure and Wiki

A useful distinction was identified between normal repository files and GitHub platform features. Files under .github/, such as issue templates and workflows, are version-controlled repository files and therefore appear after cloning. Issues, milestones, labels, and project boards are GitHub metadata and do not appear as normal local files. GitHub Wiki is special: it is associated with the repository in the web interface but is maintained as a separate Git repository.

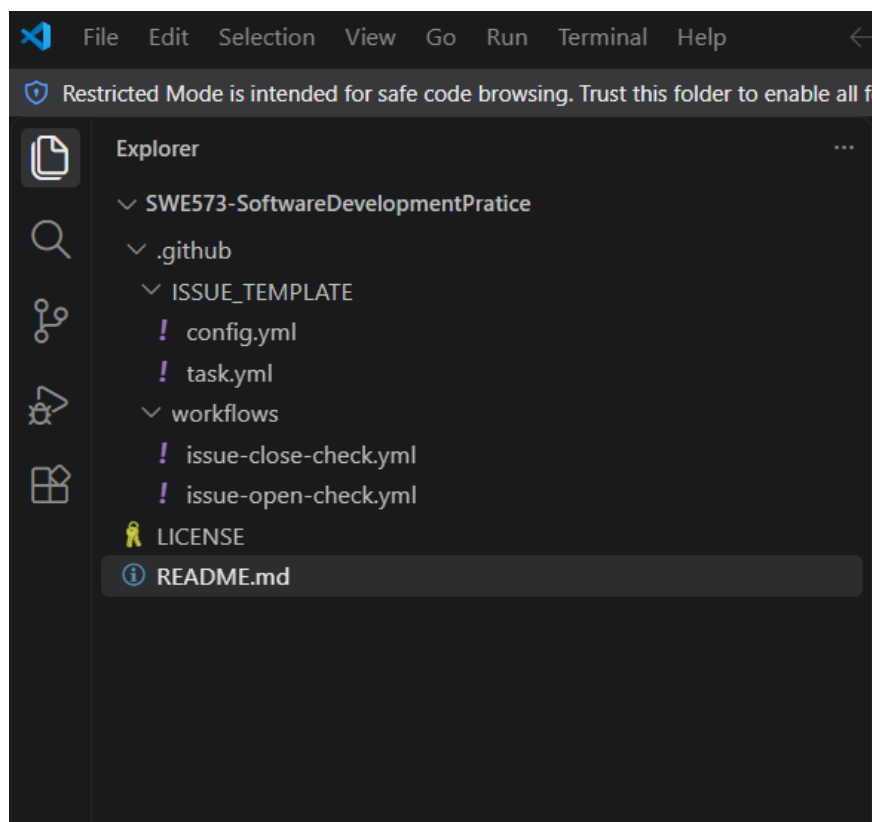


Figure 1. Example repository structure showing a version-controlled .github directory.

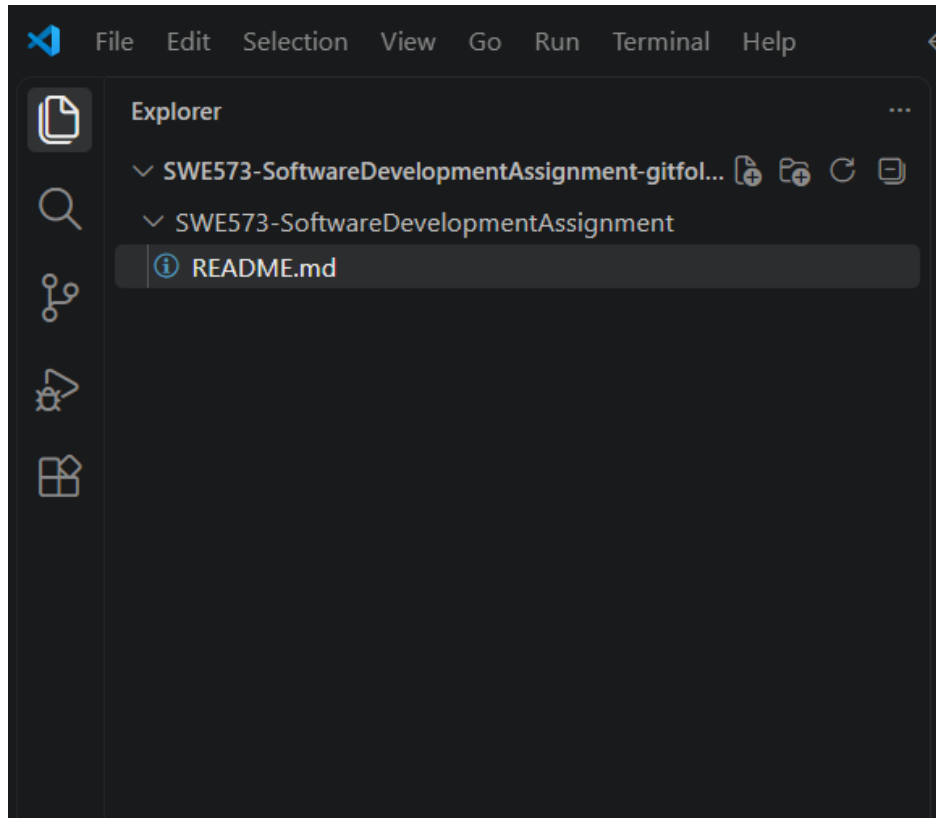


Figure 2. The local SWE573 repository at the beginning of the exercise, containing only README.md.

5. Issue labels

The repository's default labels were reviewed instead of blindly accepting another project's label scheme. A custom research label was created for investigation/comparison work. A priority family was also introduced: priority: high, priority: medium, and priority: low. High priority uses a strong red, medium uses yellow, and low uses gray so priority can be recognized visually.

The label design separates type from priority. For example, an issue can simultaneously be classified as research and priority: high. This avoids using a single label to represent two different dimensions.

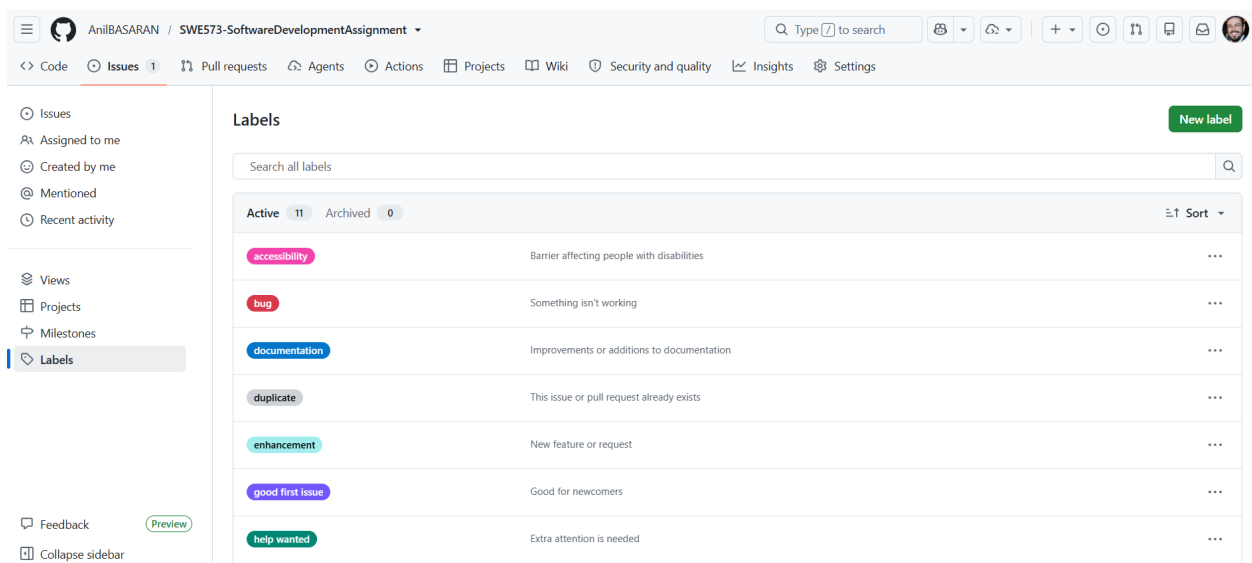


Figure 3. GitHub Labels page used to review the repository's existing labels.

Figure 4. Creating a new custom label.

6. Issue #1 - Define the application concept

The first issue was refined from a vague task about identifying project domains into [TASK]: Define the application concept. The course domain was already indigenous crafts, so the real task was to decide what application should be built inside that domain.

Acceptance criteria covered exploring possible concepts, identifying the problem and target users, selecting one concept, and documenting the selected concept in README. The issue was assigned to the student, placed under the Project Planning milestone, and tagged research plus priority: high.

Alternatives considered included an informational/blog site and a general map-based application. The selected direction was Local Craft Discovery Guide because it provides a narrower, purpose-built discovery experience rather than trying to compete directly with a general-purpose service such as Google Maps.

[TASK]: Define the application concept #1

New issue

Open

AnilBASARAN opened 2d ago

Last edited by AnilBASARAN

Owner

The general project domain is indigenous and handmade crafts.

The purpose of this task is to explore possible application ideas within this domain and decide what kind of application will be developed for the course project.

Acceptance Criteria

☐ Possible application ideas within the indigenous crafts domain are explored.

☐ The main problem that the application will address is identified.

☐ Target users are identified.

☐ One application concept is selected.

☐ The selected application concept is documented in the README.

Create sub-issue

Assignees

AnilBASARAN

Assign to Agent

Labels

priority: high

Projects

No projects

Milestone

Project Planning

No due date

Relationships

None yet

Figure 5. Issue #1 after refining its title, description, acceptance criteria, labels, assignee, and milestone.

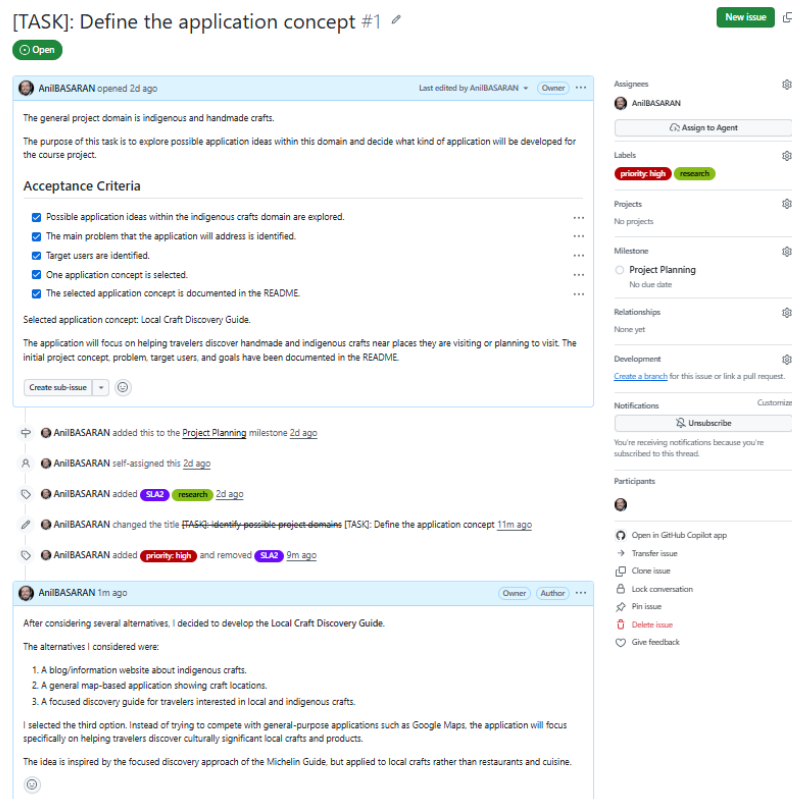


Figure 6. Completed Issue #1 with all acceptance criteria satisfied and a decision comment documenting the selected application concept.

7. Issue #2 - Prepare a potential user survey

After selecting the application concept, the next high-priority research task was requirements elicitation. Issue #2 was created as [TASK]: Prepare a potential user survey for requirements elicitation. The intention is to avoid inventing all features from assumptions and instead gather input from potential users.

The survey was structured around four questions: Who is the potential user? Why would they use the application? Where/how would they access it? What features should be prioritized?

Draft questions cover travel frequency, interest in local crafts, current discovery sources, difficulty finding authentic crafts, usage timing, preferred device, feature prioritization, the single most important feature, and open-ended feedback.

The target group was defined as people who travel regularly and enjoy exploring local culture, including people interested in discovering handmade products or purchasing meaningful souvenirs for themselves or loved ones.

[TASK]: Prepare a potential user survey for requirements elicitation #2

New issue

Open

AnilBASARAN opened 1m ago

Owner

The purpose of this task is to prepare a short survey for potential users of the Local Craft Discovery Guide. The survey will help identify user expectations, preferred platforms, and the features users consider most important. The results will later be used to prioritize the initial requirements and features of the application.

Acceptance Criteria / Requirements

- ☐ Survey objectives are defined.
- ☐ The target group for the survey is defined.
- ☐ Questions about preferred platform/device are included.
- ☐ Questions for identifying and prioritizing important features are included.
- ☐ At least one open-ended question is included to collect user suggestions.

Create sub-issue

Assignees

AnilBASARAN

Assign to Agent

Labels

priority: high research

Projects

No projects

Milestone

Project Planning
No due date

Relationships

None yet

Figure 7. Issue #2 created with high-priority research labels and measurable acceptance criteria.

[TASK]: Prepare a potential user survey for requirements elicitation #2

New issue

Open

AnilBASARAN opened 5m ago

Owner

The purpose of this task is to prepare a short survey for potential users of the Local Craft Discovery Guide. The survey will help identify user expectations, preferred platforms, and the features users consider most important. The results will later be used to prioritize the initial requirements and features of the application.

Acceptance Criteria / Requirements

- ☐ Survey objectives are defined.
- ☐ The target group for the survey is defined.
- ☐ Questions about preferred platform/device are included.
- ☐ Questions for identifying and prioritizing important features are included.
- ☐ At least one open-ended question is included to collect user suggestions.

Create sub-issue

AnilBASARAN added this to the Project Planning milestone 5m ago

AnilBASARAN self-assigned this 5m ago

AnilBASARAN added priority: high 5m ago

AnilBASARAN 1m ago

Owner

Draft Survey Questions

Travel and Local Craft Discovery

1. How often do you travel for leisure?

- Several times a year
- Once or twice a year
- Less than once a year
- Rarely / Never

2. When you travel, how often do you look for local, handmade, or indigenous crafts and products?

- Always
- Often
- Sometimes
- Rarely
- Never

3. When looking for local or handmade crafts while traveling, which sources do you usually use?

- Google Search
- Google Maps
- Instagram / Social Media
- Travel blogs or websites
- Recommendations from locals
- Travel guidebooks
- Recommendations from friends or family
- I usually discover them by chance
- Other

4. How easy or difficult is it for you to find authentic local crafts when visiting an unfamiliar destination?

- Very easy
- Easy
- Neutral
- Difficult
- Very difficult

Assignees

AnilBASARAN

Assign to Agent

Labels

priority: high research

Projects

No projects

Milestone

Project Planning
No due date

Relationships

None yet

Development

Create a branch for this issue or link a pull request

Notifications

Unsubscribe

Participants

Open to GitHub Copilot app

Transfer issue

Close issue

Lock conversation

Pin issue

Delete issue

Star feedback

Figure 8. Draft survey questions documented as an issue comment, preserving the work-in-progress reasoning in GitHub.

8. Current project-management flow

The emerging project flow is intentionally incremental:

#1 Define application concept -> #2 Prepare user survey -> #3 Conduct survey -> #4 Analyze results -> #5 Define initial requirements -> MVP / implementation

This keeps issues focused on meaningful deliverables. For example, preparing a survey is separate from conducting it and separate again from analyzing the results.

9. Key learning outcomes from this iteration

The most important learning was the difference between the working directory, staging area, local commit history, and remote branch. Commit hashes were understood as identifiers for specific commits; HEAD, main, and origin/main were understood as references rather than commit IDs themselves. The exercise also clarified the difference between Git repository content and GitHub platform metadata, and showed how issues, labels, milestones, acceptance criteria, and comments can preserve both project decisions and the reasoning behind them.

Recommended repository use

This PDF can be stored in the repository under a documentation folder, for example docs/SWE573_Project_Development_Log_Iteration_1.pdf. It should complement, not replace, the Wiki. The Wiki is better for living Git notes and command examples, while this PDF acts as a dated snapshot of the project's development and learning process.

Next planned steps

The next iteration can conduct the potential-user survey, document responses, analyze feature priorities, and convert the findings into initial software requirements. Git Wiki pages can also document the Git commands and outputs learned during this iteration using the student's own examples.